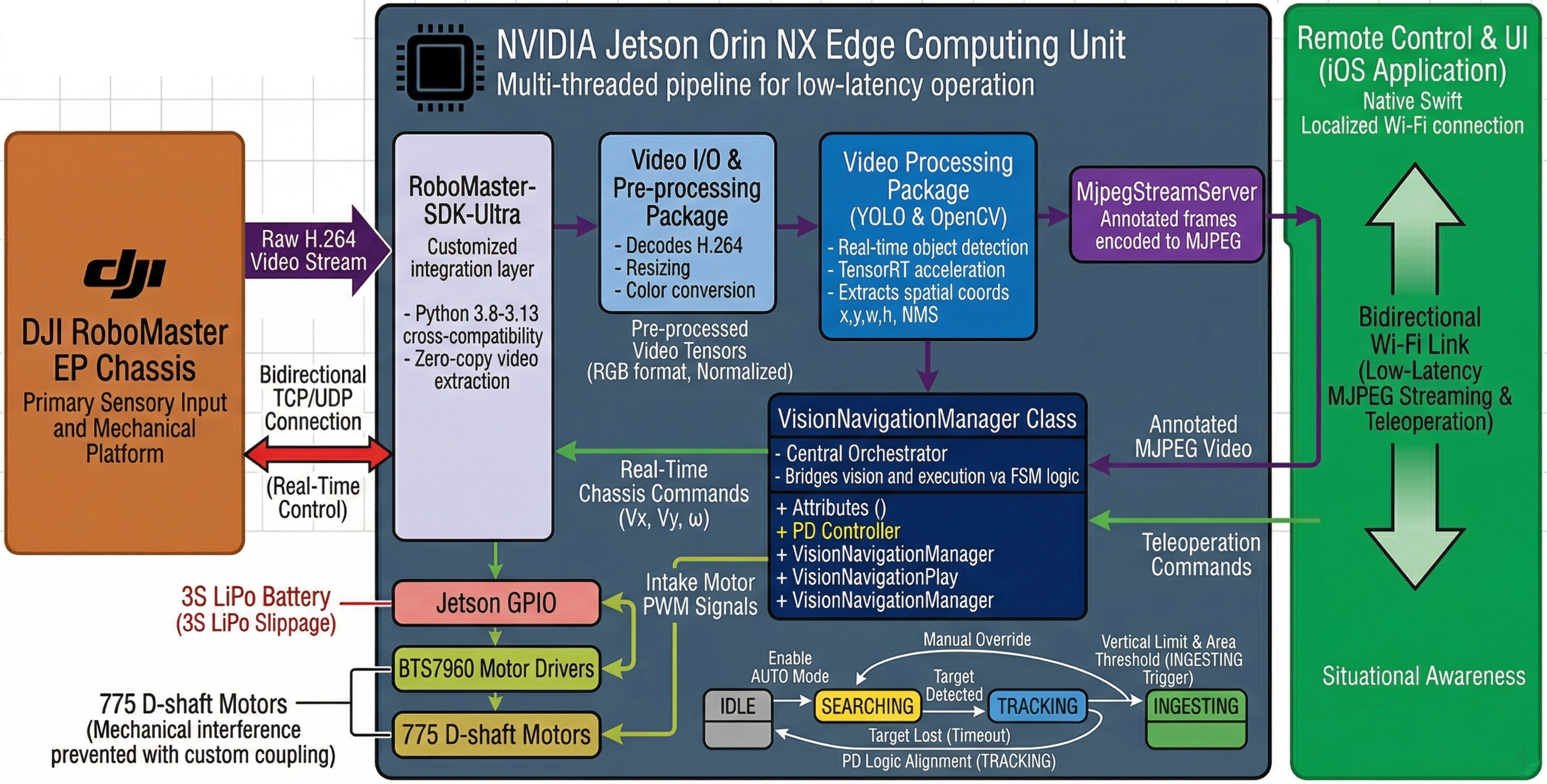
NVIDIA Jetson Orin NX Edge Computing Unit
Multi-threaded pipeline for low-latency operation
DJI RoboMaster EP Chassis
Primary Sensory Input and Mechanical Platform
Raw H.264 Video Stream
Bidirectional TCP/UDP Connection
(Real-Time Control)
RoboMaster-SDK-Ultra
Customized integration layer
- Python 3.8-3.13 cross-compatibility
- Zero-copy video extraction
Video I/O & Pre-processing Package
- Decodes H.264
- Resizing
- Color conversion
Pre-processed Video Tensors (RGB format, Normalized)
Video Processing Package (YOLO & OpenCV)
- Real-time object detection
- TensorRT acceleration
- Extracts spatial coords x,y,w,h, NMS
MjpegStreamServer
Annotated frames encoded to MJPEG
Remote Control & UI (iOS Application)
Native Swift
Localized Wi-Fi connection
Bidirectional Wi-Fi Link (Low-Latency MJPEG Streaming & Teleoperation)
Situational Awareness
VisionNavigationManager Class
- Central Orchestrator
- Bridges vision and execution va FSM logic
+ Attributes ()
+ PD Controller
+ VisionNavigationManager
+ VisionNavigationPlay
+ VisionNavigationManager
Annotated MJPEG Video
Teleoperation Commands
Real-Time Chassis Commands (Vx, Vy, ω)
Intake Motor PWM Signals
3S LiPo Battery (3S LiPo Slippage)
Jetson GPIO
BTS7960 Motor Drivers
775 D-shaft Motors
775 D-shaft Motors (Mechanical interference prevented with custom coupling)
Manual Override
Enable AUTO Mode
Target Detected
Vertical Limit & Area Threshold (INGESTING Trigger)
IDLE
SEARCHING
TRACKING
INGESTING
Target Lost (Timeout)
PD Logic Alignment (TRACKING)